Supplementary Appendix

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This appendix has been provided by the authors to give readers additional information about the work.

Supplementary Appendix

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Supplementary methods

Trial design

The study protocol originally mandated that human epidermal growth factor receptor 2 (HER2) status be determined centrally but was later amended to allow local health or regulatory authority-approved tests for local HER2 status determination.

A tumor biopsy was obtained after progression on a trastuzumab-based regimen to confirm HER2 status after trastuzumab-based therapy. Biopsies could be from either primary or metastatic sites depending on clinical context. Rebiopsy after initial negative HER2 result was not allowed.

Prophylactic or supportive treatment of study-drug-induced adverse events were permitted per investigator's discretion and institutional guidelines. As trastuzumab deruxtecan (T-DXd) is emetogenic and may cause delayed nausea and/or vomiting, patients should be premedicated with a combination regimen of 2 or 3 medicinal products (eg, dexamethasone with either a 5-hydroxytryptamine receptor antagonist and/or a neurokinin-1 receptor antagonist, as well as other medicinal products as indicated) before each dose of T-DXd to prevent nausea and vomiting.

Inclusion criteria

- Adults according to local regulation
- Pathologically documented gastric cancer and gastroesophageal junction adenocarcinoma (GEJA) that has been previously treated in the metastatic setting (unresectable, locally advanced, or metastatic disease)

- Progression on or after first-line therapy with a trastuzumab or approved trastuzumab biosimilar–containing regimen
    - Prior neoadjuvant or adjuvant therapy with a trastuzumab-containing regimen could be counted as a line of therapy if the patient progressed on or within 6 months of completing neoadjuvant or adjuvant therapy
    - Prior neoadjuvant or adjuvant therapy that did not include trastuzumab was not counted as a line of therapy, regardless of the progression status of the patient
- Locally or centrally confirmed HER2-positive (immunohistochemistry [IHC] 3+ or IHC 2+ or in situ hybridization–positive [ISH+]) as classified by American Society of Clinical Oncology and College of American Pathologists on a tumor biopsy obtained after progression on or after a first-line trastuzumab or approved trastuzumab biosimilar–containing regimen
- Eastern Cooperative Oncology Group performance status of 0 or 1 at screening
- Adequate laboratory parameters within 14 days of randomization:
    - Adequate bone marrow function
    - Adequate renal function
    - Adequate hepatic function
    - Adequate blood clotting function
- Left ventricular ejection fraction (LVEF) ≥50% within 28 days before randomization per echocardiogram or multigated acquisition scan
- Recovered from the effects of any prior surgery or radiotherapy

- Adequate treatment washout period before randomization defined as:

Treatment	**Washout Period**
Major surgery	≥4 weeks
Radiation therapy, including palliative stereotactic radiation to chest	≥4 weeks
Palliative stereotactic radiation therapy to other anatomic areas	≥2 weeks
Anticancer (immunotherapy [non-antibody-based therapy]), retinoid therapy	≥3 weeks
Targeted agents and small molecules	≥2 weeks or 5 half-lives, whichever is longer
Nitrosoureas or mitomycin C	≥6 weeks
Antibody-based anticancer therapy	≥4 weeks
Chloroquine/hydroxychloroquine	>14 days
Cell-free and concentrated ascites reinfusion therapy, peritoneal shunt or drainage of pleural effusion, ascites or pericardial effusion	≥2 weeks prior to the screening assessment

Exclusion criteria

- Use of anticancer therapy after trastuzumab-containing treatment
- Medical history of myocardial infarction within 6 months before randomization or enrollment, or symptomatic congestive heart failure (New York Heart Association Class 2 to 4).
- QT interval corrected by Fridericia's formula (QTcF) prolongation to >470 msec (female participants) or >450 msec (male participants) based on average of the screening triplicate 12-lead electrocardiogram
- History of (noninfectious) interstitial lung disease or pneumonitis that required steroids, current interstitial lung disease or pneumonitis, or where suspected interstitial lung disease or pneumonitis cannot be ruled out by imaging at screening

- Lung-specific intercurrent clinically significant illnesses including, but not limited to, any underlying pulmonary disease
- Any autoimmune, connective tissue, or inflammatory disorders where there was documented, or a suspicion of, pulmonary involvement at the time of screening
- Prior complete pneumonectomy
- Spinal cord compression or clinically active central nervous system metastases, defined as untreated and symptomatic or requiring therapy with corticosteroids or anticonvulsants to control associated symptoms.
    - Patients with clinically inactive brain metastases and patients with previously treated nonsymptomatic brain metastases were allowed
    - Patients who required no treatment with corticosteroids or anticonvulsants and had recovered from the acute toxic effect of radiotherapy were eligible to be included in the study. A minimum of 2 weeks must have elapsed between the end of whole-brain radiotherapy and randomization or study enrollment
- Patient had multiple primary malignancies within 3 years, except adequately resected nonmelanoma skin cancer, curatively treated in situ disease, other curatively treated solid tumors
- History of severe hypersensitivity reactions to trastuzumab deruxtecan or its inactive ingredients
- History of severe hypersensitivity reactions to other monoclonal antibodies
- Known allergy or hypersensitivity to paclitaxel or any components used in the paclitaxel preparation or other contraindication for taxane therapy

- Current uncontrolled infection requiring antibiotics, antivirals, or antifungals or an unexplained fever >38.0°C during screening visits or on the first scheduled day of dosing
- Substance abuse or any other medical conditions such as clinically significant cardiac or pulmonary diseases or psychological conditions
- Known human immunodeficiency virus infection or active hepatitis B or C infection, such as those with serologic evidence of viral infection within 28 days of cycle 1 day 1
- Unresolved toxicities from previous anticancer therapy, defined as toxicities (other than alopecia) not yet resolved to grade ≤1 or baseline. Patients could have enrolled with chronic, stable, grade 2 toxicities (defined as not worsening to grade >2 for at least 3 months before randomization and managed with standard-of-care treatment)
- Prior treatment with an antibody–drug conjugate consisting of an exatecan derivative that is a topoisomerase I inhibitor
- Pregnant, breastfeeding, or planning to become pregnant
- Clinically significant gastrointestinal disorder (including hepatic disorders, bleeding, inflammation, occlusion, ileus, diarrhea grade >1, jaundice, intestinal paralysis, malabsorption syndrome, ulcerative colitis, inflammatory bowel disease, or partial bowel obstruction)
- History of receiving live, attenuated vaccine within 30 days before the first exposure to study intervention.

Endpoints

- Overall survival (OS), defined as the time from date of randomization until death from any cause

- Progression-free survival (PFS), defined as the time from date of randomization until first objective radiographic tumor progression or death from any cause
- Confirmed objective response rate (ORR), defined as the proportion of patients who achieve a best response of complete response or partial response using the Response Evaluation Criteria in Solid Tumours, version 1.1 (confirmation of complete and partial responses were required by a subsequent assessment)
- Disease control rate, defined as the proportion of patients who achieved complete response, partial response, or stable disease for a minimum of 6 weeks during study treatment
- Duration of response (DoR), defined as time from the initial response (complete response or partial response) until documented tumor progression or death from any cause and based on investigator's assessment

Patient-Reported Outcomes

Health Economics and Outcomes Research based on patient-reported outcomes were used to evaluate study treatment. The impact of gastric cancer on symptoms and quality of life were assessed based upon the European Organisation for Research and Treatment of Cancer (EORTC) EuroQol 5-Dimension 5-Level (EQ-5D-5L) visual analog scale (VAS) and the Functional Assessment of Cancer Therapy–Gastric (FACT-Ga) gastric cancer subscales

OS Sensitivity Analyses

A rank-preserving structural failure time model to adjust the OS for patients in the ramucirumab plus paclitaxel group who received T-DXd or disitamab vedotin after they discontinued the study drug was assessed. OS censored for subsequent anti-cancer therapy after study drug

discontinuation and OS in patients with centrally confirmed HER2-positive IHC status were also assessed.

Subgroup Analyses

Subgroups analyzed included the stratification factors HER2 status, geography, and time to progression on first-line therapy. Additional subgroups included age, region, Eastern Cooperative Oncology Group performance status, primary tumor location, histological subtype, number of metastatic sites, and prior treatment with immune checkpoint inhibitors.

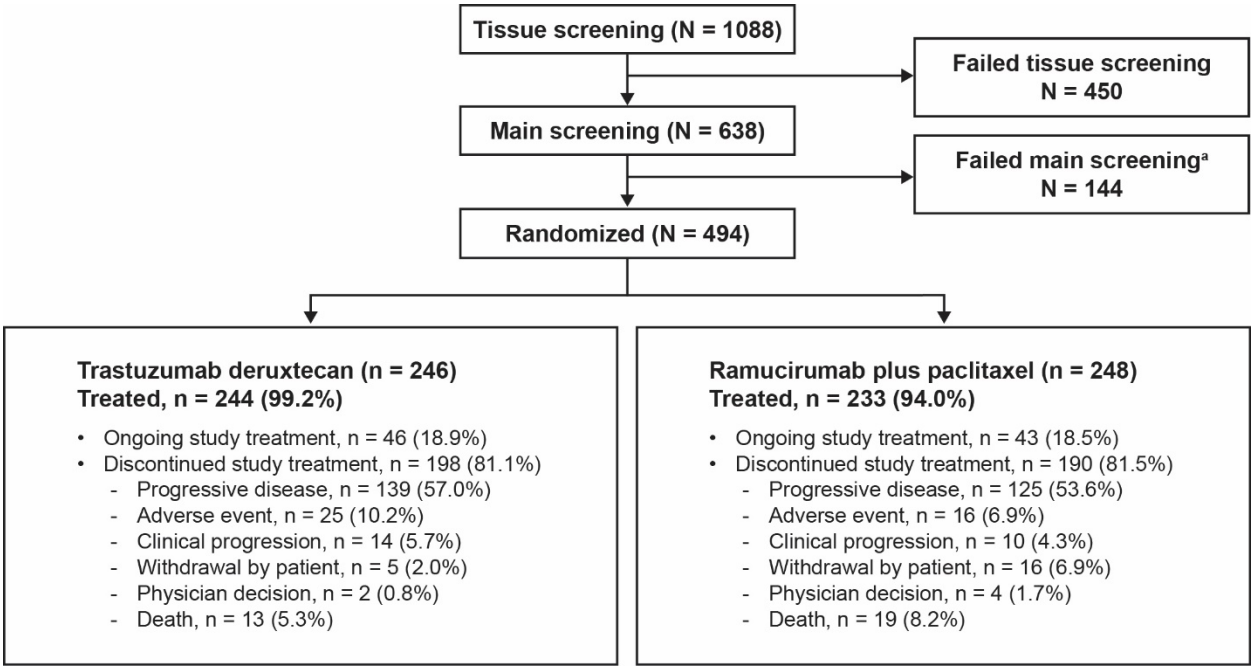
Statistical Analyses

The Kaplan-Meier method was applied to estimate the survival distribution of OS for each treatment arm while a stratified log-rank test was used to compare the distribution of PFS between the treatment arms. Kaplan-Meier estimates of OS and PFS rates at fixed time points and the 2-sided 95% CI was also provided. Median OS and PFS time with 95% CI for each arm was calculated using the Brookmeyer-Crowley method. The stratified Cox proportional hazards regression model was used to calculate the hazard ratios and 95% CIs of OS and PFS, using the same stratification factors as randomization stratification factors. Confirmed ORRs were summarized by treatment arm, along with their 2-sided 95% CIs using Clopper-Pearson methods. Cochran-Mantel-Haenszel tests, also stratified by the randomization stratification factors, were used to compare confirmed ORR between the treatment arms. DoR and its 2-sided 95% CI were summarized for each treatment group using the Brookmeyer-Crowley method.

Descriptive statistics of safety endpoints were conducted. The safety analysis was performed on the safety analysis set, which included all patients who received at least 1 dose of the study drug.

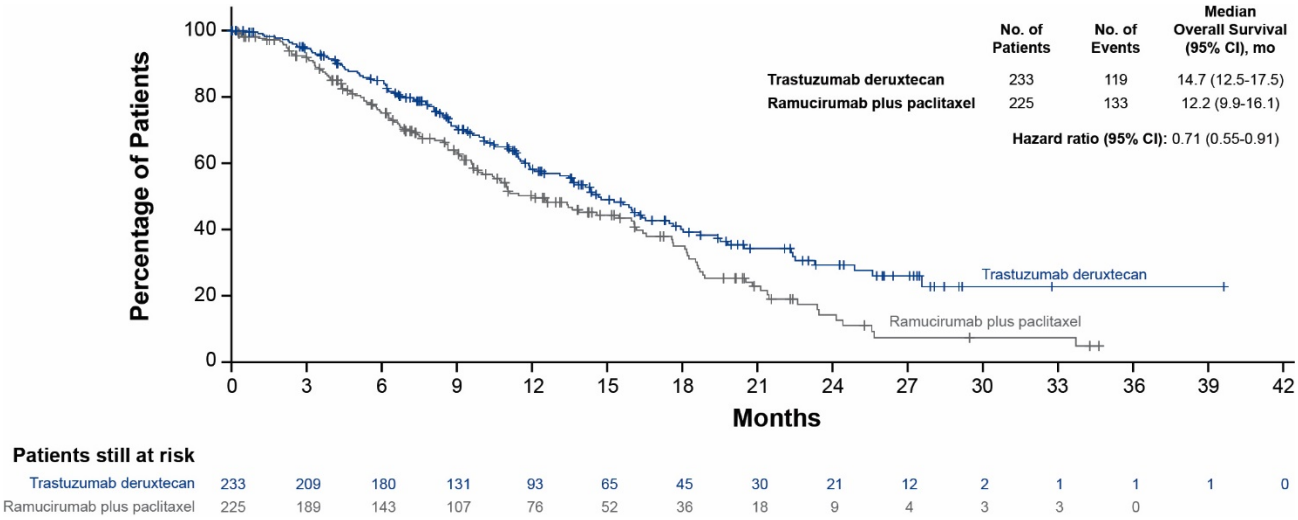
Supplementary Figures and Tables

Figure S1. Patient Disposition.



[a]Reasons for screen failure were not satisfying inclusion/exclusion criteria (n = 129), withdrawal by patient (n = 9), adverse event (n = 2), physician decision (n = 1), and other (n = 3).

Figure S2. Overall Survival Based on Patients with Centrally Tested HER2-Positive Status.



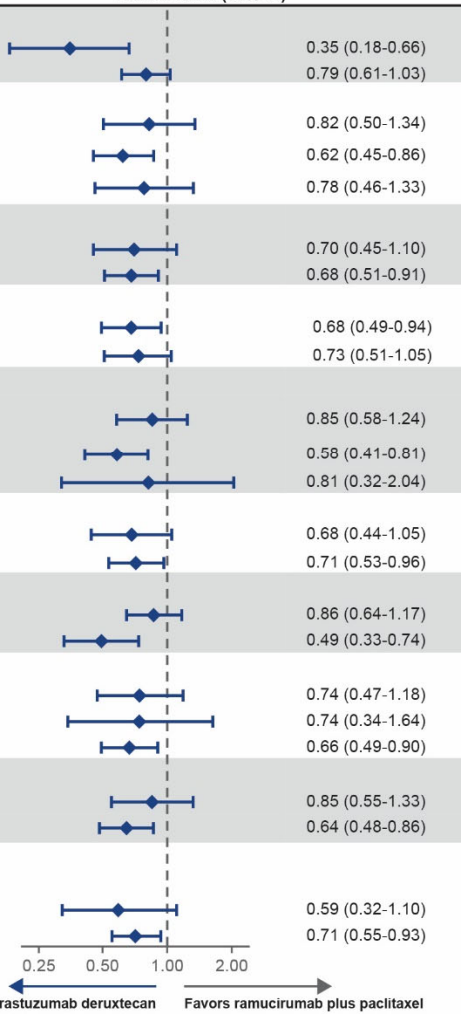
HER2, human epidermal growth factor receptor 2; IHC, immunohistochemistry; ISH, in situ hybridization; OS, overall survival; T-DXd, trastuzumab deruxtecan.

Of the 494 patients included in the study, 458 patients had centrally tested HER2-positive results (central testing for eligibility or central analysis of local testing used for eligibility). Patients with centrally tested HER2-positive status showed an OS hazard ratio for death of 0.71 (95% CI, 0.55 to 0.91) which is consistent with results of the OS from the full analysis set. In this analysis, 119 (51.1%) and 133 (59.1%) OS events were observed with T-DXd and ramucirumab plus paclitaxel, respectively; other patients had their data censored (tick marks).

Figure S3. Subgroup Analysis of Overall Survival (A) and Progression-Free Survival (B).

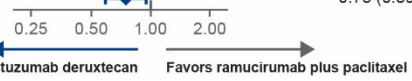
A

	No. of Events/No. of Patients		OS, median (95% CI), mo		Hazard Ratio (95% CI)
	Trastuzumab deruxtecan	Ramucirumab plus paclitaxel	Trastuzumab deruxtecan	Ramucirumab plus paclitaxel	
HER2 IHC status					
IHC 2+/ISH+	17/39	25/40	17.7 (9.1-NE)	7.4 (4.5-10.7)	0.35 (0.18-0.66)
IHC 3+	107/207	117/208	14.4 (11.9-16.4)	13.4 (10.8-16.5)	0.79 (0.61-1.03)
Geography					
Asia	31/57	34/60	14.4 (10.5-19.4)	16.1 (10.8-18.9)	0.82 (0.50-1.34)
Western Europe	67/140	79/139	15.7 (11.9-18.8)	11.1 (8.7-15.5)	0.62 (0.45-0.86)
Mainland China/Rest of world	26/49	29/49	14.8 (8.7-NE)	9.7 (7.4-16.0)	0.78 (0.46-1.33)
Time to progression on first line					
<6 months	38/61	38/61	10.2 (8.1-14.4)	7.3 (5.0-10.1)	0.70 (0.45-1.10)
≥6 months	86/185	104/187	16.3 (13.7-19.4)	13.5 (11.0-17.6)	0.68 (0.51-0.91)
Age					
<65 years	69/140	75/130	15.7 (12.1-18.1)	11.1 (9.6-16.1)	0.68 (0.49-0.94)
≥65 years	55/106	67/118	13.7 (10.5-17.5)	11.5 (8.9-16.6)	0.73 (0.51-1.05)
Region					
Asia (including mainland China)	52/97	54/95	14.4 (11.5-18.1)	14.6 (9.7-18.2)	0.85 (0.58-1.24)
Europe	60/123	80/140	15.9 (12.5-19.5)	11.0 (8.6-13.9)	0.58 (0.41-0.81)
Rest of world	12/26	8/13	7.8 (4.7-22.4)	12.2 (3.0-17.6)	0.81 (0.32-2.04)
ECOG PS					
0	38/97	45/88	18.1 (13.1-27.6)	17.6 (12.6-20.6)	0.68 (0.44-1.05)
1 or 2	86/149	96/159	13.9 (10.2-16.0)	9.9 (7.7-11.4)	0.71 (0.53-0.96)
Primary tumor location					
Gastric	83/153	83/149	11.9 (10.1-14.8)	11.4 (9.6-16.1)	0.86 (0.64-1.17)
GEJ	41/93	59/99	18.0 (14.7-22.5)	11.5 (9.2-16.2)	0.49 (0.33-0.74)
Histological subtype					
Intestinal	35/69	38/66	16.6 (11.5-20.5)	12.6 (9.9-18.3)	0.74 (0.47-1.18)
Diffuse	15/25	11/19	9.4 (7.9-16.0)	8.7 (4.6-16.1)	0.74 (0.34-1.64)
Others	74/152	93/163	14.7 (11.9-19.4)	11.1 (9.2-16.1)	0.66 (0.49-0.90)
Number of metastatic sites					
<2	36/73	44/75	13.6 (10.1-22.4)	15.5 (10.8-17.6)	0.85 (0.55-1.33)
≥2	88/173	98/173	14.7 (11.9-17.7)	10.4 (8.7-13.5)	0.64 (0.48-0.86)
Prior treatment with immune checkpoint inhibitors					
Yes	18/39	24/38	16.3 (10.2-NE)	10.1 (7.4-18.6)	0.59 (0.32-1.10)
No	106/207	118/210	14.4 (11.9-17.5)	12.6 (9.9-16.1)	0.71 (0.55-0.93)



B

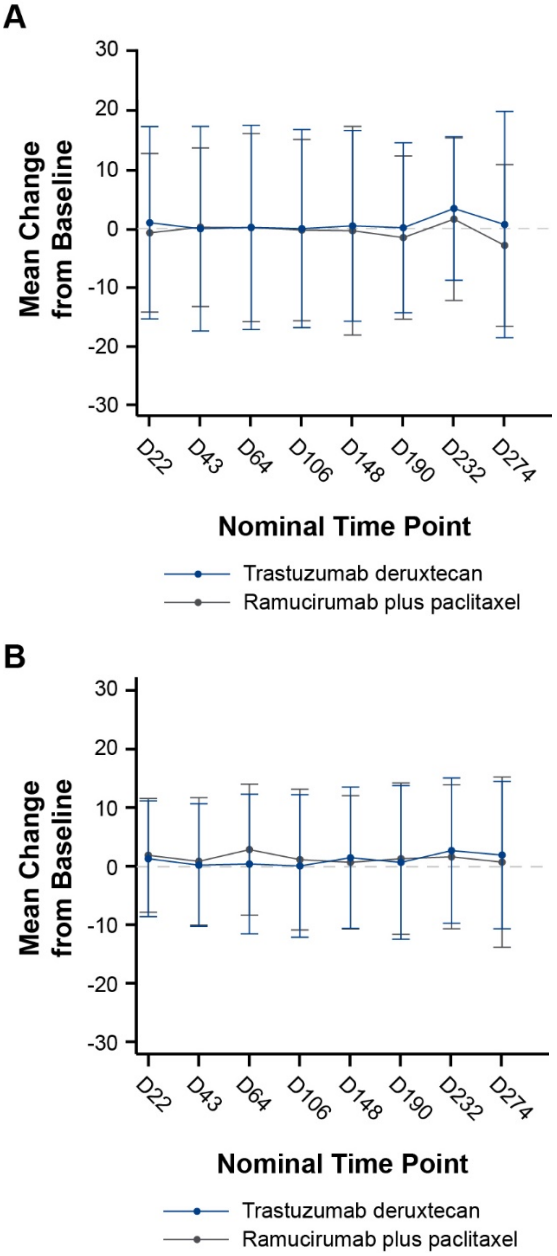
	No. of Events/No. of Patients		PFS, median (95% CI), mo		Hazard Ratio (95% CI)
	Trastuzumab deruxtecan	Ramucirumab plus paclitaxel	Trastuzumab deruxtecan	Ramucirumab plus paclitaxel	
HER2 IHC status					
IHC 2+/ISH+	27/39	27/40	4.4 (3.0-9.7)	3.6 (2.4-5.4)	0.64 (0.37-1.11)
IHC 3+	139/207	129/208	6.7 (5.6-7.1)	5.6 (5.1-6.4)	0.76 (0.60-0.97)
Geography					
Asia	39/57	37/60	6.9 (4.4-10.9)	5.4 (4.0-8.1)	0.74 (0.47-1.16)
Western Europe	94/140	83/139	5.9 (5.6-6.9)	5.6 (4.8-5.9)	0.77 (0.57-1.04)
Mainland China/Rest of world	33/49	36/49	7.2 (4.2-9.6)	5.6 (4.4-7.0)	0.62 (0.38-1.00)
Time to progression on first line					
<6 months	49/61	41/61	5.4 (4.0-5.7)	4.2 (2.8-5.0)	0.78 (0.51-1.18)
≥6 months	117/185	115/187	6.9 (5.9-8.1)	5.6 (5.4-6.6)	0.72 (0.55-0.93)
Age					
<65 years	96/140	86/130	6.8 (5.7-7.5)	5.6 (4.2-6.4)	0.75 (0.56-1.00)
≥65 years	70/106	70/118	5.7 (5.3-7.8)	5.6 (4.8-6.5)	0.72 (0.52-1.01)
Region					
Asia (including mainland China)	66/97	63/95	7.0 (5.4-8.4)	5.6 (4.2-7.0)	0.72 (0.51-1.02)
Europe	84/123	87/140	6.7 (5.6-7.1)	5.5 (4.4-5.8)	0.72 (0.54-0.98)
Rest of world	16/26	6/13	5.6 (3.2-7.8)	5.7 (0.7-NE)	0.76 (0.29-2.01)
ECOG PS					
0	63/97	43/88	6.9 (5.6-8.3)	6.0 (4.4-7.9)	0.80 (0.54-1.19)
1 or 2	103/149	113/159	6.3 (5.1-7.0)	5.5 (4.4-5.7)	0.74 (0.57-0.97)
Primary tumor location					
Gastric	102/153	86/149	5.9 (5.1-7.4)	5.6 (4.4-6.8)	0.83 (0.62-1.10)
GEJ	64/93	70/99	6.9 (5.7-8.0)	5.5 (4.4-5.8)	0.62 (0.44-0.88)
Histological subtype					
Intestinal	47/69	36/66	6.3 (5.1-8.7)	4.9 (4.2-5.6)	0.75 (0.48-1.16)
Diffuse	13/25	13/19	6.9 (5.6-14.0)	4.2 (2.6-8.0)	0.44 (0.20-1.00)
Others	106/152	107/163	6.7 (5.6-7.2)	5.6 (5.3-6.5)	0.78 (0.60-1.02)
Number of metastatic sites					
<2	43/73	57/75	6.8 (5.6-8.6)	5.8 (4.8-7.9)	0.71 (0.47-1.05)
≥2	123/173	99/173	6.7 (5.6-7.1)	5.4 (4.4-5.6)	0.74 (0.57-0.97)
Prior treatment with immune checkpoint inhibitors					
Yes	28/39	27/38	7.2 (4.4-9.6)	5.4 (4.2-8.1)	0.74 (0.43-1.27)
No	138/207	129/210	6.4 (5.6-7.0)	5.6 (4.8-5.8)	0.75 (0.59-0.95)



ECOG PS, Eastern Cooperative Oncology Group performance status; GEJ, gastroesophageal junction; HER2, human epidermal growth factor receptor 2; IHC, immunohistochemistry; ISH, in situ hybridization; OS, overall survival; PFS, progression-free survival.

Panel A shows hazard ratios and 95% CIs for median OS in subgroups according to patients and tumor characteristics. The OS benefit of T-DXd over ramucirumab plus paclitaxel was consistent across most of the subgroups. Panel B shows hazard ratios and 95% CIs for PFS in subgroups defined according to patients and tumor characteristics. The benefit of T-DXd over ramucirumab plus paclitaxel was consistent across most of the subgroups.

Figure S4. EQ-5D-5L VAS (A) and FACT-Ga Gastric Cancer Subscale (B).



EQ-5D-5L, EuroQol 5 dimension, 5 level; FACT-Ga, Functional Assessment of Cancer Therapy–Gastric; T-DXd, trastuzumab deruxtecan; VAS, visual analog scale.

Scores remained stable with no clinically meaningful changes in both arms during study treatment. A clinically meaningful change is defined as a ≥10-point change from baseline. Baseline completion for EQ-5D-5L VAS (panel A) was 98.2% (n = 217) for T-DXd and 99.5% (n = 203) for ramucirumab plus paclitaxel. Baseline completion for FACT-Ga Gastric Cancer Subscale (panel B) was 99.1% (n = 219) for T-DXd and 97.5% (n = 199) for ramucirumab plus paclitaxel. The completion rate was calculated as number of patients who had PRO assessment values at the specified visit) / (number of patients who had the specified visit number) x 100.

Table S1. Representativeness of Study Participants.

Disease under investigation	**Gastric cancer**
Special considerations related to:	
Sex	Gastric cancer incidence and mortality are higher in men than in women regardless of race and ethnicity.[1-3] In 2022, the age-standardized incidence rate was 12.8% and 6.0% in men and women, respectively. The age-standardized death rate was 8.6% in men and 3.9% in women.[1]
Age	The majority (>80%) of gastric cancer cases occur in patients aged 60 years or older.[2,4] However, an increase in incidence of gastric cancer in younger age groups has been noted in recent years.[1,2]
Race/ethnicity	Based on cancer statistics in the United States between 2017 and 2021, American Indian/Alaska Natives have the highest incidence rates of gastric cancer, with rates up to 10.3 per 100,000, followed by African Americans (9.9 per 100,000), Hispanic/Latinos (9.4 per 100,000), and Asian American/Pacific Islanders (8.9 per 100,000). Whites have the lowest incidence rates with 5.2 per 100,000.[5]
Geography	The highest incidence and mortality rates of gastric cancer are reported in Asia.[2,3,6] The top 5 countries with new cases of gastric cancer reported in 2022 by age standardized rates included Mongolia, Japan, South Korea, Iran, and Tajikistan.[7]
Other considerations	Approximately 5%-17% of gastric cancers have HER2 IHC 3+ or IHC 2+/ISH+ expression.[8-10]
Overall representativeness of this study	The demographic and epidemiologic characteristics of the DESTINY-Gastric04 population were generally comparable to available data for patients with HER2-positive unresectable/metastatic gastric cancer or GEJA.[1,4-6] Of the 494 patients randomly assigned to receive either T-DXd or ramucirumab plus paclitaxel, most were men (76.0% and 82.7%, respectively). Median age of patients enrolled in the trial was 63.2 years in the T-DXd arm and 64.3 years in the ramucirumab plus paclitaxel arm. Racial diversity was limited, with most patients being White (47.2% and

	52.4%) or Asian (41.1% and 39.1%) and very few patients being Black or African American (0% and 0.8%). The majority of randomized patients had a HER2 score of IHC 3+ compared with IHC 2+/ISH+ (84.1% vs 15.9% for the T-DXd group and 83.9% vs 16.1% for the RAM + PTX group); this is in line with previous trials studying HER2-positive gastric cancer.[11-13]

Table S2. Additional Overall Survival Analyses.

	T-DXd n = 246	**Ramucirumab Plus Paclitaxel n = 248**
OS based on centrally confirmed HER2-positive status, median (95% CI),[a,b] months	14.7 (12.5-17.5)	12.2 (9.9-16.1)
HR (95% CI)[c]	0.71 (0.55-0.91)	
OS censored for subsequent anticancer therapy after study drug discontinuation,[d] median (95% CI),[a] months	NE (18.8-NE)	15.5 (10.8-NE)
HR (95% CI)[e]	0.64 (0.44-0.93)	
RPSFT-adjusted OS,[d] median (95% CI),[a] months	14.7 (12.1-16.6)	11.0 (9.4-14.2)
HR (95% CI)[f]	0.67 (0.49-0.89)	

HER2, human epidermal growth factor receptor 2; HR, hazard ratio; NE, not estimable; OS, overall survival; RPSFT, rank-preserving structural failure time; T-DXd, trastuzumab deruxtecan.

[a]Median OS and OS rates at specified timepoints were calculated from Kaplan-Meier analysis. CI for median was computed using the Brookmeyer-Crowley method.

[b]233 patients in the T-DXd group and 225 patients in the ramucirumab plus paclitaxel group had centrally confirmed HER2-positive results (central testing for eligibility or central analysis of local testing used for eligibility).

[c]Stratified Cox proportional hazards model was adjusted for the HER2 status (IHC 3+ or IHC 2+/ISH+) stratification factor.

[d]A pre-defined sensitivity analysis which censored posttrial anticancer therapy demonstrated an OS hazard ratio for death of 0.64 (95% CI, 0.44 to 0.93). Furthermore, a rank-preserving structural failure time model, used to adjust the OS of patients in the ramucirumab plus paclitaxel group who received posttrial T-DXd or disitamab vedotin showed an OS hazard ratio for death of 0.67 (95% CI, 0.49 to 0.89). These results were consistent with the primary analysis favoring T-DXd (Table 2).

[e]Unstratified Cox proportional hazards models are used when there are less than 5 events in any of the treatment groups per stratum.

[f]95% CI was estimated using bootstrapping percentile method with seed 123.

Table S3. Posttrial Anticancer Therapy.[a]

no. (%)	T-DXd n = 246	Ramucirumab Plus Paclitaxel n = 248
BDC 1001	0	1 (0.4)
Bevacizumab	0	1 (0.4)
Brucea javanica oil	0	1 (0.4)
Calcineurin subunit B	0	1 (0.4)
Calcium folinate	5 (2.0)	2 (0.8)
Calcium folinate; fluorouracil; irinotecan	1 (0.4)	0 (0.0)
Calcium folinate; fluorouracil; irinotecan hydrochloride	11 (4.5)	12 (4.8)
Capecitabine	7 (2.8)	0
Capecitabine; cisplatin	1 (0.4)	0
Capecitabine; irinotecan	0	1 (0.4)
Capecitabine; oxaliplatin	1 (0.4)	1 (0.4)
Carboplatin	1 (0.4)	0
Cisplatin	1 (0.4)	2 (0.8)
Combinations of antineoplastic agents	1 (0.4)	0
Disitamab vedotin	5 (2.0)	12 (4.8)
Docetaxel	4 (1.6)	4 (1.6)
Fluorouracil	10 (4.1)	11 (4.4)
Fluorouracil sodium	1 (0.4)	0
Fluorouracil; folinic acid; oxaliplatin	2 (0.8)	3 (1.2)
Fluorouracil; irinotecan	0	1 (0.4)
Fluorouracil; oxaliplatin	1 (0.4)	0
Folinic acid	2 (0.8)	4 (1.6)
Fruquintinib	3 (1.2)	1 (0.4)
Gimeracil; oteracil potassium; tegafur	3 (1.2)	0
GS 1811	0	1 (0.4)
Immunotherapy	1 (0.4)	1 (0.4)
Investigational drug	5 (2.0)	10 (4.0)
Irinotecan	16 (6.5)	16 (6.5)
Irinotecan hydrochloride	2 (0.8)	7 (2.8)
Irinotecan hydrochloride trihydrate	1 (0.4)	0
Irinotecan hydrochloride trihydrate liposomal	1 (0.4)	0
Levofolinic acid	1 (0.4)	0
Levonorgestrel	0	1 (0.4)
Marsdenia tenacissima stem	0	1 (0.4)
Nivolumab	14 (5.7)	14 (5.6)
Oxaliplatin	10 (4.1)	1 (0.4)

Oxyfedrine hydrochloride	0	1 (0.4)
Paclitaxel	63 (25.6)	8 (3.2)
Paclitaxel nanoparticle albumin-bound	13 (5.3)	0
Paclitaxel; ramucirumab	13 (5.3)	3 (1.2)
Pembrolizumab	2 (0.8)	2 (0.8)
Pyrotinib maleate	1 (0.4)	1 (0.4)
Ramucirumab	43 (17.5)	8 (3.2)
Regorafenib	0	1 (0.4)
Rivoceranib	1 (0.4)	1 (0.4)
Rivoceranib mesylate	1 (0.4)	2 (0.8)
Runimotamab	0	1 (0.4)
Serplulimab	0	1 (0.4)
Sintilimab	5 (2.0)	2 (0.8)
Sulfasalazine	0	1 (0.4)
Tegafur	0	1 (0.4)
Tipiracil	1 (0.4)	0
Tipiracil hydrochloride; trifluridine	11 (4.5)	11 (4.4)
Tipiracil; trifluridine	2 (0.8)	1 (0.4)
Tislelizumab	0	1 (0.4)
Toripalimab	0	1 (0.4)
Trastuzumab	12 (4.9)	8 (3.2)
Trastuzumab deruxtecan	3 (1.2)	52 (21.0)
Trastuzumab mafodotin	0	1 (0.4)
Trifluridine	2 (0.8)	2 (0.8)
Tumor necrosis factor NOS	1 (0.4)	0
Zimberelimab	0	1 (0.4)

T-DXd, trastuzumab deruxtecan.

[a]Patients may have been treated with more than 1 type of posttrial anticancer therapy.

Table S4. Posttrial HER2-directed Therapies by Region.[a]

Region	T-DXd n = 246 n/N[b] (%)	Ramucirumab Plus Paclitaxel n = 248 n/N[b] (%)
HER2-directed therapies by randomization stratification factor region		
Asia (excluding mainland China)		
Disitamab vedotin	0	0
T-DXd	1/57 (1.8)	25/60 (41.7)
Western Europe		
Disitamab vedotin	0	0
T-DXd	2/140 (1.4)	25/139 (18.0)
Mainland China/Rest of world		
Disitamab vedotin	5/49 (10.2)	12/49 (24.5)
T-DXd	0	2/49 (4.1)
HER2-directed therapies by region		
Asia (including mainland China)		
Disitamab vedotin	5/97 (5.2)	12/95 (12.6)
T-DXd	1/97 (1.0)	27/95 (28.4)
Europe		
Disitamab vedotin	0	0
T-DXd	2/123 (1.6)	25/140 (17.9)
Rest of world		
Disitamab vedotin	0	0
T-DXd	0	0

HER2, human epidermal growth factor receptor 2; T-DXd, trastuzumab deruxtecan.

[a]Patients who received the agent in any subsequent lines are counted into that agent. If a patient received multiple agents, the patient is counted into each agent received.

[b]N is the number of patients in each region.

Table S5. Overall Safety Summary.

	Safety Analysis Set	
no. (%)	**T-DXd n = 244**	**Ramucirumab Plus Paclitaxel n = 233**
Any-grade TEAEs Drug-related	244 (100.0) 227 (93.0)	228 (97.9) 213 (91.4)
Grade ≥3 TEAEs Drug-related	166 (68.0) 122 (50.0)	172 (73.8) 126 (54.1)
Serious TEAEs Drug-related	100 (41.0) 45 (18.4)	101 (43.3) 41 (17.6)
TEAEs associated with dose discontinuation Drug-related	35 (14.3) 28 (11.5)	40 (17.2) 31 (13.3)
TEAEs associated with dose reductions Drug-related	77 (31.6) 76 (31.1)	87 (37.3) 84 (36.1)
TEAEs associated with dose interruptions Drug-related	137 (56.1) 94 (38.5)	141 (60.5) 119 (51.1)
TEAEs associated with deaths Drug-related	22 (9.0) 4 (1.6)	35 (15.0) 2 (0.9)
Total on-treatment deaths	21 (8.6)	34 (14.6)

T-DXd, trastuzumab deruxtecan; TEAEs, treatment-emergent adverse events.

Table S6. Adverse Events of Special Interest.[a]

	Grade 1	Grade 2	Grade 3	Grade 4	Grade 5	Any Grade
Adjudicated drug-related interstitial lung disease/pneumonitis,[b] no. (%)						
T-DXd n = 244	7 (2.9)	26 (10.7)	1 (0.4)	0	0	34 (13.9)
Ramucirumab plus paclitaxel n = 233	0	0	2 (0.9)	0	1 (0.4)	3 (1.3)
Left ventricular dysfunction,[c] no. (%)						
T-DXd n = 244	0	3 (1.2)	3 (1.2)	0	0	6 (2.5)
Ramucirumab plus paclitaxel n = 233	2 (0.9)	2 (0.9)	0	0	0	4 (1.7)

AESI, adverse event of special interest; CTCAE, Common Terminology Criteria for Adverse Events; TEAE, treatment-emergent adverse event; T-DXd, trastuzumab deruxtecan.

[a]If a patient had both missing and non-missing CTCAE grades for a TEAE, the missing CTCAE grade was treated as the lowest severity grade. If a patient had more than 1 event per AESI category (or preferred term) level, the patient was counted once at each level of summation.

[b]Includes preferred terms interstitial lung disease, pneumonitis, alveolitis, and pulmonary toxicity.

[c]Includes preferred terms ejection fraction decreased, cardiac failure, cardiac failure acute, cardiac failure congestive, left ventricular dysfunction.

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